

Домашняя работа по дифференциальным уравнениям
 (N 118) Решить уравнение.

$$y' = 2 \left(\frac{y+2}{x+y-1} \right)^2$$

$$\begin{cases} x = u+3 \\ y = v-2 \end{cases}$$

$$\frac{dv}{du} = 2 \frac{v^2}{(u+v)^2}$$

$$\frac{dy}{dx} = \frac{dv}{du}$$

$$v = \varepsilon(u)u$$

$$u \frac{d\varepsilon}{du} = 2 \frac{\varepsilon^2 u^2}{u^2 (1+\varepsilon)^2} = 2 \frac{\varepsilon^2}{(1+\varepsilon)^2} - \varepsilon$$

$$u d\varepsilon = 2 \frac{\varepsilon^2 - \varepsilon(1+\varepsilon)}{(1+\varepsilon)^2} du$$

$$\frac{1}{\varepsilon(1+\varepsilon)^2} d\varepsilon = \frac{du}{u} = \frac{1}{\frac{2\varepsilon^2 - \varepsilon(1+\varepsilon)}{(1+\varepsilon)^2}} d\varepsilon$$

$$\int \frac{1}{\varepsilon(1+\varepsilon)^2} d\varepsilon = \int \frac{du}{u} = - \left(\int \frac{1}{\varepsilon} d\varepsilon + \int \frac{2}{1+\varepsilon^2} d\varepsilon \right)$$

$$-\ln \varepsilon - 2 \operatorname{arctg} \varepsilon = \ln u + C$$

$$-2 \operatorname{arctg} \varepsilon = \ln u + \ln \varepsilon + \ln C$$

$$-2 \operatorname{arctg} \varepsilon = \ln u \varepsilon C \quad -2 \operatorname{arctg} \frac{y+2}{x-3} = \ln(y+2)x$$

$$e^{-2 \operatorname{arctg} \frac{y+2}{x-3}} = C(y+2)$$

Ответ: $e^{-2 \operatorname{arctg} \frac{y+2}{x-3}} = C(y+2)$

(№121) $x^3(y-x)=y^2$ — является однородным уравнением

Замени: $y = f(x)^m \Rightarrow f = \sqrt[m]{y} = \sqrt{y}$

$$x^3(m f^{m-1} f' - x) = f^{2m}$$

Пусть $m=2$

$x^3(2f \cdot f' - x) = f^4$ — однородное уравнение:

Замени: $f = X S$, $f' = S'X + S$ $S = \frac{f}{X}$

$$x^3(2XS \cdot (S'X + S) - X) = X^4 S^4$$

$$2X^4 S (S'X + S) - X^4 = X^4 S^4$$

$$X^4(2S(S'X + S) - 1 - S^4) = 0$$

$$2S \cdot S'X + 2S^2 - 1 - S^4 = 0$$

$$2S \frac{dS}{dX} X + 2S^2 - 1 - S^4 = 0$$

$$2SX \frac{dS}{dX} = S^4 - 2S^2 + 1 = (S^2 - 1)^2$$

$$2SX dS$$

$$\frac{2S dS}{(S^2 - 1)^2}$$

$$\int \frac{2S dS}{(S^2 - 1)^2}$$

$$\begin{cases} S^2 - 1 = t \\ dt = 2S dS \end{cases}$$

$$dS =$$

Ответ

$$2s x ds = (s^2 - 1)^2 dx$$

$$\frac{2s ds}{(s^2 - 1)^2} = \frac{dx}{x}$$

$$\int \frac{2s ds}{(s^2 - 1)^2} = \int \frac{dx}{x}$$

$$\left\{ \begin{array}{l} s^2 - 1 = t \\ dt = 2s ds \\ ds = \frac{dt}{2s} \end{array} \right.$$

$$\int \frac{dt}{t^2} = \int \frac{dx}{x}$$

$$-\frac{1}{t} = \ln |cx|$$

$$-\frac{1}{s^2 - 1} = \ln |cx|$$

$$-\frac{1}{\frac{y}{x^2} - 1} = \ln |cx|$$

$$\frac{-1}{\frac{y}{x^2} - 1}$$

$$-\frac{1}{\frac{y}{x^2} - 1} = \ln |cx|$$

$$-1 = \ln |cx| \left(\frac{y}{x^2} - 1 \right)$$

$$1 = \ln |cx| \left(1 - \frac{y}{x^2} \right) = \ln |cx| \left(\frac{x^2 - y}{x^2} \right)$$

$$x^2 = \ln |cx| (x^2 - y)$$

Answer: $x^2 = \ln |cx| (x^2 - y); y = x^2$

to check solution:
let $y = x^2$

$$x^3 (2x - x) = x^4$$

$$x^3 \cdot x = x^4 - \text{correct} \Rightarrow$$

$\Rightarrow y = x^2$ - solution
remember 9.9

(N125) $2y' + x = 4\sqrt{y}$ — не является однородной

Замени: $y = f(x)^m$

$$y' = m f(x)^{m-1} f'$$

$$2(m f(x)^{m-1} f' + x) = 4 f^{\frac{m}{2}}$$

пусть $m=2$ $4 f f' + 2x = 4 f$

$$f' = \frac{df}{dx}$$

$$2 f f' + \frac{x}{2} = 2 f$$

$$2 f \frac{df}{dx} + \frac{x}{2} = 2 f$$

$$2 f \frac{df}{dx} - 2 f = -\frac{x}{2}$$

Замени: $f = tx$

$$4tx(t+x') + x = 4tx$$

$$4t^2x + 4x^2 - t^2x = 4tx$$

$$t(4x^2)' = 4tx - x - 4t^2x$$

$$t(4x^2) \frac{dt}{dx} = x(4t - 1 - 4t^2)$$

$$t(4x^2) dt = x(4t - 1 - 4t^2) dx$$

$$\frac{4t dt}{4t - 1 - 4t^2} = \frac{x dx}{2x^2} \approx \frac{dx}{2x} = \frac{4t dt}{-(2t-1)^2}$$

$$\int \frac{4+t}{(2t-1)^2} = - \int \frac{dx}{2x}$$

$$2 \left(\int \frac{(2t-1) dt}{(2t-1)^2} + \int \frac{dt}{(2t-1)^2} \right) = \ln|2t-1| - \frac{1}{2t-1}$$

$$\ln|2t-1| - \frac{1}{2t-1} = -\frac{\ln x}{2} + C$$

$$\ln|2t-1| - \frac{1}{2t-1} = -\ln Cx$$

$$t = \frac{f}{x} \quad \ln \left| 2 \frac{f}{x} - 1 \right| - \frac{1}{2 \frac{f}{x} - 1} = -\ln Cx$$

$$f = \sqrt{y} \quad \ln \left| \frac{2\sqrt{y} - x}{x} \right| - \frac{x}{2\sqrt{y} - x} = -\ln Cx$$

$$\ln |(2\sqrt{y} - x)C| = \frac{x}{2\sqrt{y} - x}$$

при $2\sqrt{y} - x = 0$ $2\sqrt{y} = x$ — закрое перемен.

Пробуем: ~~2x~~

$$\sqrt{y} = \frac{x}{2}$$

$$y = \frac{x^2}{4}$$

$$dy = \frac{2x}{4} = \frac{x}{2}$$

$$2 \cdot \frac{x}{2} + x = 4 \cdot \frac{x}{2}$$

$$2x = 2x - \text{вын.}$$

$$\text{Ответ: } \ln |(2\sqrt{y} - x)C| = \frac{x}{2\sqrt{y} - x} \quad \sqrt{y} = \frac{x}{2}.$$

N146

$$(2e^y - X)y' = 1$$

$$\frac{dy}{dx}(2e^y - X) = \frac{1}{y'}$$

$$\frac{dy}{dx} = \frac{1}{\frac{dy}{dx}}$$

$$2e^y - X = \frac{1}{y'} = \frac{1}{\frac{dy}{dx}} = X'$$

$$2e^y = X' + X$$

Найти частные и общие решения:

$$1) X' + X = 0$$

$$\frac{dX}{dy} + X = 0$$

$$dX + X dy = 0$$

$$X dy = -dX$$

$$dy = -\frac{dX}{X}$$

$$y = -\ln|X| + C$$

$$X = Ce^{-y} \quad (1)$$

$$C = C(y)$$

$$2e^y = Ce^{-y} + e' e^{-y} \cdot e^{-y}$$

$$2e^y = C' e^{-y}$$

$$C' = \frac{2e^y}{e^{-y}} = 2e^{2y}$$

$$C(y) = e^{2y} + \tilde{C} \quad (2)$$

Подставляем (2) в (1)

X =

Ответ:

N152

при X =

или

$$X = (e^{2y} + \tilde{C})e^{-y} = e^y + e^{-y}\tilde{C}$$

ответ: $X = e^y + e^{-y}\tilde{C}$

(N152) $(X+1)(y' + y^2) = -y$

при $X \neq -1$ $y' + y^2 = \frac{-y}{X+1}$

$$y' + \frac{y}{X+1} = -y^2$$

Обозначим лев. часть уравнения Бернулли.

$$y' + p(x)y = f(x) \cdot y^n, \quad n \neq 0, n \neq 1 \quad | : y^n$$

$$f(x) = \frac{1}{y} = p(x)(X+1)$$

$$\frac{y'}{y^2} + \frac{1}{y(X+1)} = -1$$

$$y = \frac{1}{f(x)}$$

$$-\frac{f'}{f^2} \cdot f^2 + \frac{f}{(X+1)} = -1$$

$$y' = -\frac{1}{f^2(x)}$$

$$-f' + \frac{f}{X+1} = -1$$

~~$$e^{f(x)}(X+1) = f$$~~

$$df = 1 - \frac{f}{X+1} dx$$

~~$$e^{f(x)}(X+1) = \frac{1}{y}$$~~

$$-f'(X+1) + f = -1(X+1)$$

~~$$\frac{f'-1}{f} = X+1=0 \text{ при } X=-1$$~~

$$f'(X+1) - f = X+1$$

$$(f'-1)(X+1) = f \Rightarrow f = C(X)(X+1)$$

~~$$\frac{df}{dx} = 1 \quad f = X + C(X) \quad f' = 1 + C'(X)$$~~

$$\frac{1}{y} = C(x)(x+1) \quad y = \frac{1}{C(x)(x+1)}$$

$$y' + \frac{y}{x+1} = -y^2 \quad \text{т.е. при } y=0$$

$0+0=0 \Rightarrow \text{реш.}$

$$- \frac{C'(x)(x+1) + C(x)}{C^2(x)(x+1)^2} = - \frac{1}{C^2(x)(x+1)^2}$$

$$-(C'(x)(x+1) + C(x)) = -1$$

$$-C'(x)(x+1) = -1$$

$$C'(x)(x+1) = 1$$

$$C'(x) = \frac{1}{x+1}$$

$$C(x) = \int \frac{1}{x+1} dx = \ln |\tilde{C}(x+1)|$$

$$\frac{1}{y} = \ln |\tilde{C}(x+1)| (x+1)$$

$$y = \frac{1}{\ln |\tilde{C}(x+1)| (x+1)}$$

Ответ: $y = \frac{1}{\ln |\tilde{C}(x+1)| (x+1)} \quad , y=0$

(N154)

Задание

(N154) $xy^2y' = x^2 + y^3$
 $\frac{1}{3}x \frac{dy^3}{dx} = x^2 + y^3$

Заменим: $y^3 = t$ $\frac{1}{3}x t' = x^2 + t$ (1)
 $\frac{1}{3}x dt = (x^2 + t) dx$ (при $x^2 = 0$)
 $\frac{dt}{t} = \frac{dx}{\frac{1}{3}x} = \frac{3dx}{x}$

$$\ln t = 3 \ln x + \ln C$$

$$t = Cx^3 \text{ (подставляем в (1))}$$

$$\frac{1}{3}x \cdot (3x^2C + C^3x^3) = x^2 + Cx^3$$

$$x^3C + \frac{1}{3}C^3x^4 = x^2 + Cx^3$$

$$\frac{x^4}{3}C' = x^2 \quad \text{т.к. при } x=0$$

$0 = 0 + y^3 \Rightarrow y = 0$
 т.к. решение входит в (2),
 не является решением (1.с)

$$C' = x^2: \frac{x^4}{3} = \frac{3}{x^2}$$

$$C(x) = \int \frac{3}{x^2} dx = -\frac{3}{x} + \tilde{C}$$

$$y^3 = \left(-\frac{3}{x} + \tilde{C}\right)x^3 = -3x^2 + \tilde{C}x^3$$

$$y^3 = \tilde{C}x^3 - 3x^2 \quad (2)$$

ответ: $y^3 = \tilde{C}x^3 - 3x^2$